

Example 10.5

$$\text{ARIMA}(0, 1, 1) \times (0, 1, 1)_4$$

$$\text{Period} = 4$$

$$P = 0, D = 1, Q = 1 \quad \leftarrow \text{Seasonal components}$$

$$p = 0, d = 1; q = 1 \quad \leftarrow \text{Non-seasonal components}$$

$$\phi_p(B) = 1; \quad \Phi_p(B^4) = 1$$

$$\theta_q(B) = 1 + \theta_1 B$$

$$\Theta_q(B^4) = 1 + \Theta_1 B^4$$

$$\phi_p(B) \Phi_p(B^4) (1-B)^d (1-B^4)^D X_t = \theta_q(B) \Theta_q(B^4) Z_t$$

$$(1-B)(1-B^4) X_t = (1 + \theta_1 B)(1 + \Theta_1 B^4) Z_t$$

$$(1 - B - B^4 + B^5) X_t = (1 + \theta_1 B + \Theta_1 B^4 + \theta_1 \Theta_1 B^5) Z_t$$

$$X_t - X_{t-1} - X_{t-4} + X_{t-5}$$

$$= Z_t + \theta_1 Z_{t-1} + \Theta_1 Z_{t-4} + \theta_1 \Theta_1 Z_{t-5}.$$